

Project Progress Report

To: Babilon Mobile

From: Franz Kingstein Nesakumar

Date: June 10, 2025

Subject: Success progress Report – Whisper STT and GUI Development

Overview

This report outlines the progress made on the Speech-to-Text (STT) and GUI development for the Tajik language STT system powered by OpenAI's Whisper model. The tasks span from tech stack planning to GUI integration and active Whisper model training.

Completed Tasks

Task Name	Date	Status
Document for tech stack	May 28, 2025	Done
Code for Whisper, GUI starting	May 29, 2025	Done
Completed GUI and first corrections	May 30, 2025	Done
Presented GUI to team lead	June 2, 2025	Done
First GUI documentation completed	June 6, 2025	Done

These milestones cover foundational design, Whisper setup, and iterative improvements to the GUI after early feedback.

Work in Progress

Task Name	Start Date	Status
Started with the model training	June 3, 2025	In progress
Error detection and correction of model	June 4, 2025	In progress
Re-correcting the model	June 9, 2025	In progress

Technical Status & Notes

- **Model Training**

I am currently fine-tuning the Whisper model using the **Tajik subset of Google's FLEURS dataset**.

While significant preprocessing, dataset normalization, and streaming optimizations have been implemented, I'm facing technical obstacles due to:

- Transformers version limitations (currently using 4.52.4)
 - Incompatibilities with `TrainingArguments` in this version
 - Memory constraints while batching variable-length audio features
 - Frequent interruptions due to resource limits and long training durations on local hardware (non-Colab environment)
 - **GUI Progress**
A responsive and WebSocket-ready React + Tailwind GUI has been developed, including:
 - An editable avatar placeholder
 - Functional chat with emoji support and backend-ready message emitters
 - A control panel with buttons for mic, volume, and recording—all event-ready for backend hooks
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Deliverables Prepared

- `whisper.py`: Memory-efficient model training script with checkpointing and early stopping
 - UI Codebase: Modular React components with backend integration points
 - Backend Integration Memo (for teammates): Clear instructions for how backend devs can hook into UI controls
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Current Challenge

Despite optimizing data pipelines and using streaming for low-RAM usage, model training continues to be resource-intensive and time-consuming on local systems.

Efforts are ongoing to:

- Break dataset into smaller checkpoints
 - Resume training automatically after interruptions
 - Prepare state-safe recovery logic
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Next Steps

- Resolve batching logic in collator (in testing)
 - Deploy final round of GUI integration
 - Stabilize and document backend sockets
 - Complete full evaluation and export trained Whisper model
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Thank you for your continued support. Please advise if cloud compute resources can be allocated to help expedite training.

Sincerely,

Franz Kingstein Nesakumar

Speech UI Developer, Babilon Mobile